

Q1) AR(3):

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + a_t, \quad a_t \sim N(0, \sigma_a^2) \text{ iid.}$$

obs data:

$$Y_t = X_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

$$\Rightarrow \underline{X}_t = \begin{bmatrix} X_t \\ X_{t-1} \\ X_{t-2} \\ X_{t-3} \end{bmatrix}, \quad \underline{G}_t = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \underline{W}_t = \begin{bmatrix} a_t \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \underline{V}_t = \begin{bmatrix} \sigma_a^2 & 0 & 0 \\ 0 & \sigma_\varepsilon^2 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 \end{bmatrix}$$

$$\underline{X}_t = \underline{G}_t \underline{X}_{t-1} + \underline{W}_t = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_{t-1} \\ X_{t-2} \\ X_{t-3} \end{bmatrix} + \begin{bmatrix} a_t \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow Y_t = X_t + \varepsilon_t = \underline{F}_t^T \underline{X}_t + V_t$$

$$\therefore \underline{F}_t = [1 \ 0 \ 0], \quad V_t = \varepsilon_t, \quad V_t = \sigma_\varepsilon^2$$

Q2) $Y_t = \underline{F}_t^T \underline{X}_t + V_t, \quad X_t = G_t X_{t-1} + W_t$

$$\underline{X}_t = \begin{bmatrix} d_{1t} & B_{1t} & d_{2t} & B_{2t} \end{bmatrix}^T, \quad Y_t = \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix}$$

$$\underline{X}_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} d_{1t-1} \\ B_{1t-1} \\ d_{2t-1} \\ B_{2t-1} \end{bmatrix} + \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \\ y_{4t} \end{bmatrix}$$

$$\underline{Y}_t = \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} 1 & R_{11t} & 0 & 0 \\ 0 & 0 & 1 & R_{21t} \end{bmatrix} \begin{bmatrix} d_{1t} \\ B_{1t} \\ d_{2t} \\ B_{2t} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

$$\Rightarrow \underline{G}_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \underline{W}_t = \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \\ y_{4t} \end{bmatrix}, \quad \underline{W}_t = \begin{bmatrix} \sigma_{y_1}^2 & 0 & 0 & 0 \\ 0 & \sigma_{y_2}^2 & 0 & 0 \\ 0 & 0 & \sigma_{y_3}^2 & 0 \\ 0 & 0 & 0 & \sigma_{y_4}^2 \end{bmatrix}$$

$$V_t = \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}, \quad \underline{V}_t = \begin{bmatrix} \sigma_{\varepsilon_1}^2 & r_{\varepsilon} \\ r_{\varepsilon} & \sigma_{\varepsilon_2}^2 \end{bmatrix}$$

$$\underline{F}_t = \begin{bmatrix} 1 & R_{11t} & 0 & 0 \\ 0 & 0 & 1 & R_{21t} \end{bmatrix}$$

HW6 – Ying Zheng

Problem 1

(a)

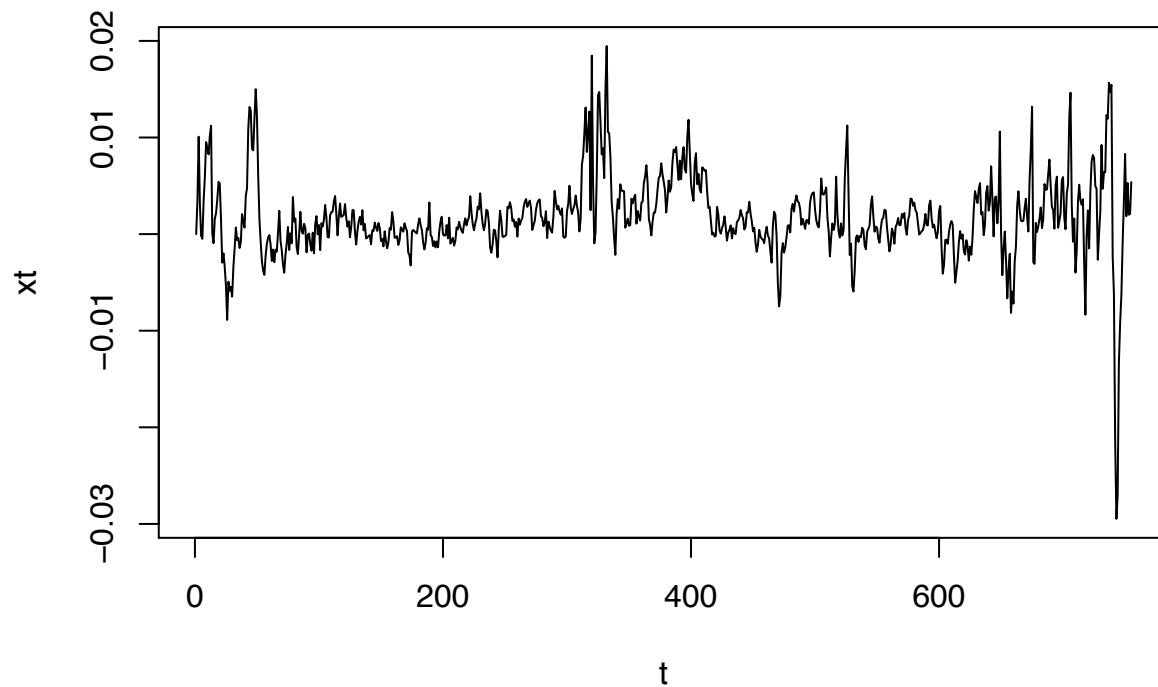
```
buildAR3 = function(x)
{
  FF = matrix(c(1,0,0),nrow=1)
  GG = matrix(c(x[1],1,0,x[2],0,1,x[3],0,0),nrow=3)
  V = exp(x[4])
  W = matrix(c(exp(x[5]),rep(0,8)),nrow=3)
  m0 = rep(0,3)
  CO = matrix(solve(diag(9) - GG %x% GG) %*% matrix(W,ncol=1),ncol=3)
  return(list(m0 = m0, CO = CO, FF = FF, GG = GG, V = V, W = W))
}

ppi= read.table('m-ppiaco4709.txt')
#Year, Month, Day, PPI
zt = diff(log(ppi$V4), 1)
yt = zt-mean(zt)

holdfit = dlmMLE(yt,c(1,1,1,1,1),buildAR3)
estimated_parameters = holdfit$par
holdfit

## $par
## [1] 1.02204084 -0.21312549 -0.01673239 -10.08680704 -11.63898805
##
## $value
## [1] -3279.631
##
## $counts
## function gradient
##      63      63
##
## $convergence
## [1] 0
##
## $message
## [1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"

filter_ = dlmFilter(zt,buildAR3(holdfit$par))
plot(ts(filter_ $m[, 1]), type = 'l',ylab = 'xt', xlab='t')
```



```
sigma.square.e = exp(-10.08680704)
sigma.square.a = exp(-11.63898805)
print(c(sigma.square.e,sigma.square.a))
```

```
## [1] 4.162511e-05 8.815597e-06
```

The fitted model is: $X_t = 1.02204084X_{t-1} - 0.21312549X_{t-2} - 0.01673239X_{t-3} + a_t$, where $a_t \sim \text{iid } N(0, 4.162511e-05)$

$Y_t = X_t + e_t$, where $e_t \sim N(0, 8.815597e-06)$